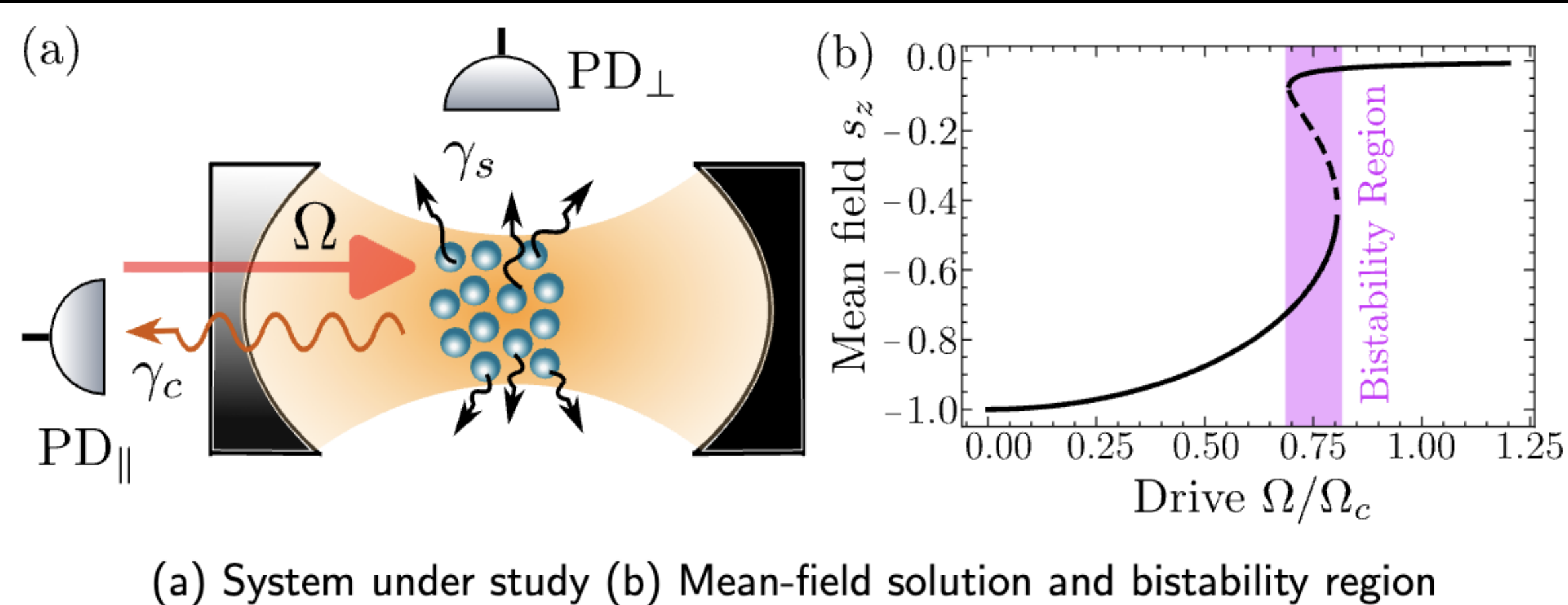


# Predicting Open Quantum Systems Trajectories: TrajectoryGRUGrid



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## Physics (short): Driven – Dissipative cavity



## Quantum Trajectories Approach

Main equation – Master equation

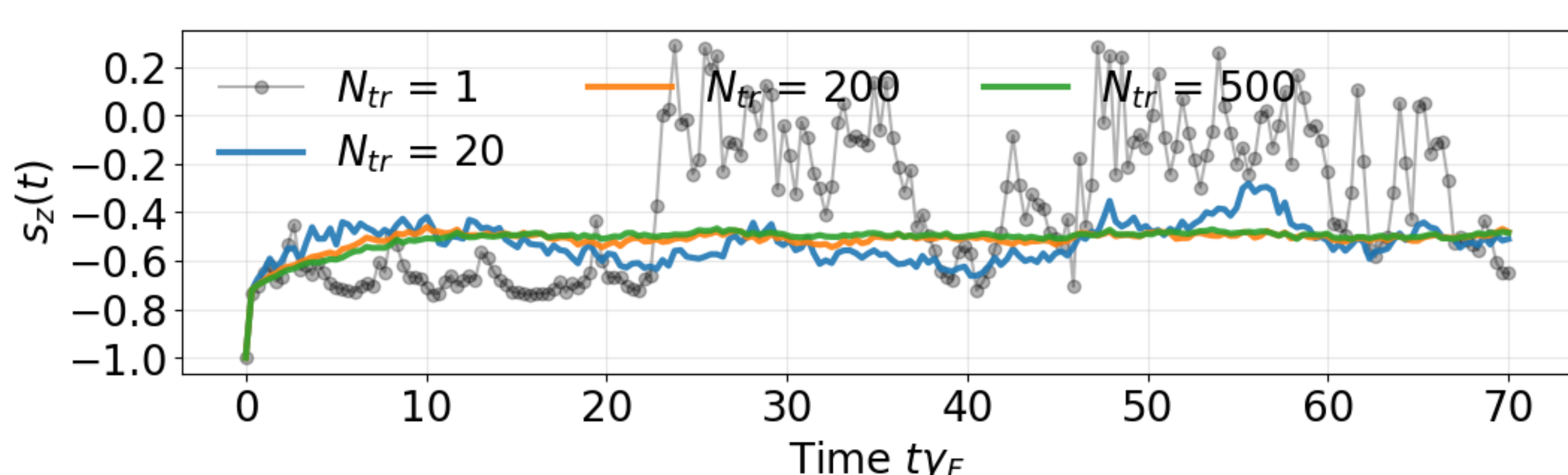
$$\dot{\rho} = -i[\mathcal{H}, \rho] + \frac{\gamma_c}{2} \mathcal{D}[\hat{S}_-]\rho + \frac{\gamma_s}{2} \sum_n \mathcal{D}[\hat{\sigma}_n]\rho, \quad \mathcal{H} = 2\Omega \hat{S}_x$$

Generally: Solve for density matrix:  $4^N \times 4^N$  elements,  $N$  is the number of qubits.  
Instead: Probabilistic unraveling of the master equation into quantum trajectories

$$|\psi(t+dt)\rangle = \begin{cases} \frac{\hat{S}_- |\psi(t)\rangle}{\sqrt{p_-}} & \text{with probability } p_- = \gamma_c dt \langle \psi(t) | \hat{S}_+ \hat{S}_- | \psi(t) \rangle \\ \frac{\hat{\sigma}_n |\psi(t)\rangle}{\sqrt{p_n}} & \text{with probability } p_n = \gamma_s dt \langle \psi(t) | \hat{\sigma}_n^\dagger \hat{\sigma}_n | \psi(t) \rangle \\ \frac{e^{-i\mathcal{H}_{NH}dt} |\psi(t)\rangle}{\sqrt{p_0}} & \text{with probability } p_0 = 1 - p_- - \sum_n p_n \end{cases}$$

✓ Saves memory:  $2^N \times 2^N$  elements

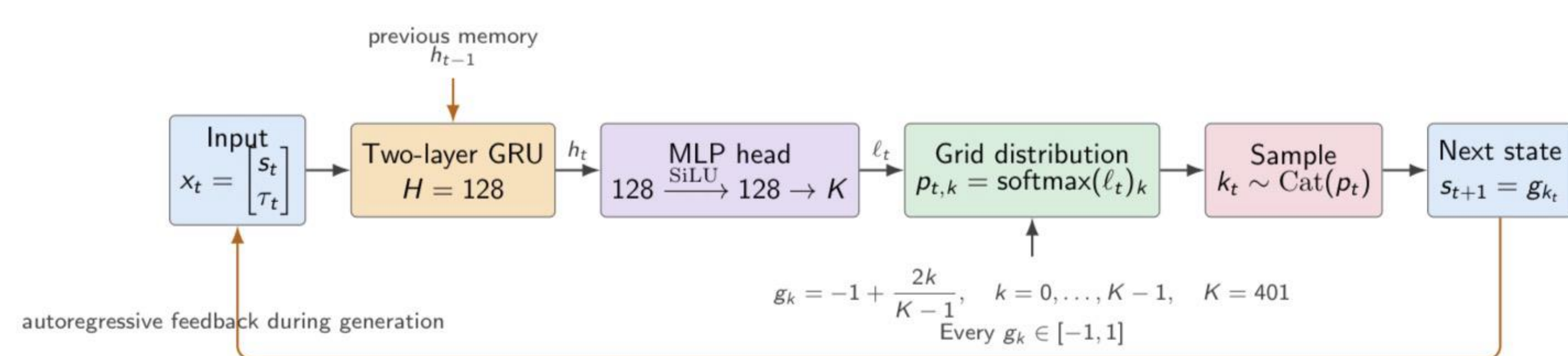
✗ Loses Time: Need to average over many trajectories to get the density matrix.



## TrajectoryGRUGrid: Model

We use a Gate Recurrent Unit (GRU) model for predicting time series on a **discrete** grid  $x_{t+1} = f_\theta(x_t|h_t)$ ;  $h_t$  - hidden state. Input:  $x_t = [s_t; \tau_t]$ ; Output:  $s_{t+1} \in [-1, 1]$

Bounded autoregressive trajectory model



Training at every time step

$$y_t = s_{t+1}^{\text{real}}, \quad q_{t,i} = 1 - \alpha, \quad q_{t,i+1} = \alpha, \quad \mathcal{L}_t = - \sum_{k=0}^{K-1} q_{t,k} \log p_{t,k}.$$

real  $s_t$  as input  $\rightarrow$  two-bin interpolated target  $q_t$   $\rightarrow$  categorical cross-entropy.

teacher forcing  $\rightarrow$  continuous data on a discrete grid  $\rightarrow$  maximize next-state likelihood

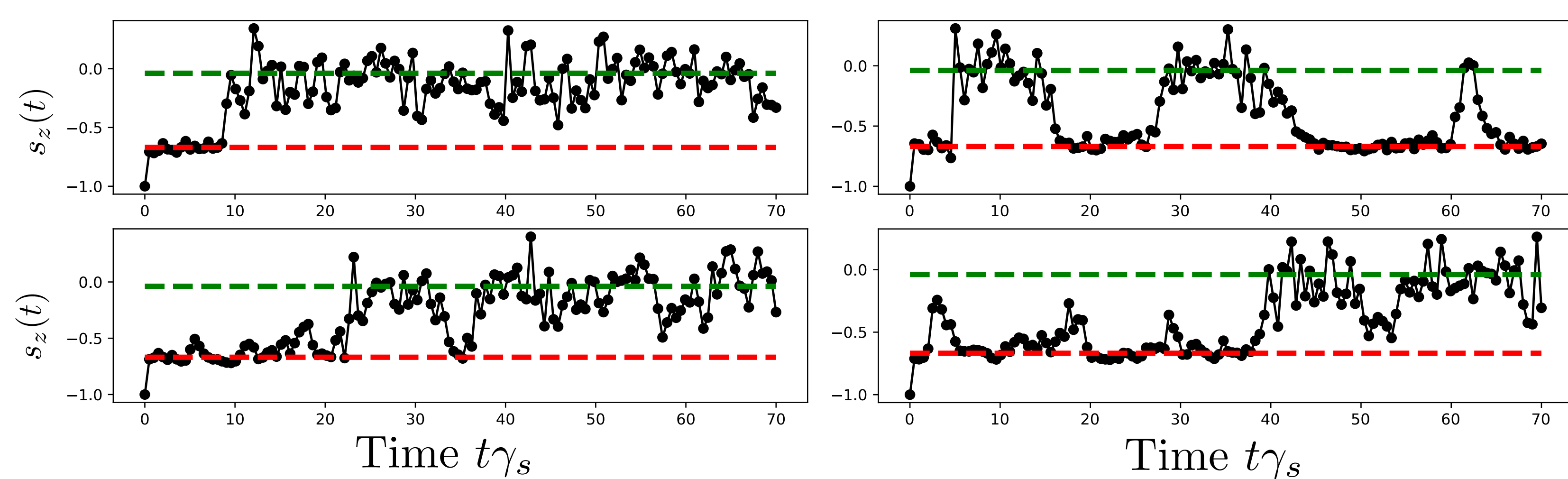
## Dataset: Quantum Trajectories with N and Ω

- Number of Atoms:  $N \in \{4, 6, 8, 10, 12, 14, 16, 18\}$
- Drive ratios:  $\Omega \in \{0.61, 0.63, 0.65, 0.67, 0.69, 0.71, 0.73, 0.75\}$
- Trajectories per condition = 500

## Conclusions

- The GRU-grid model successfully learned bounded stochastic trajectory dynamics conditioned on N and Omega.
- Model learns the parameters of the time series (bistability, switching time, etc.)
- When the model is trained, the generation of the new trajectories is instant
- It reproduced metastable branches and stochastic switching for fixed parameters.
- Despite training only up to N=18, it showed physically meaningful behavior up to N~30.
- Small errors in switching probabilities and dwell times produced deviations in ensemble averages.
- Interpolation in  $\Omega$  was reliable, while extrapolation occasionally produced a weak spurious metastable branch.
- Probably increasing the model will give even better results

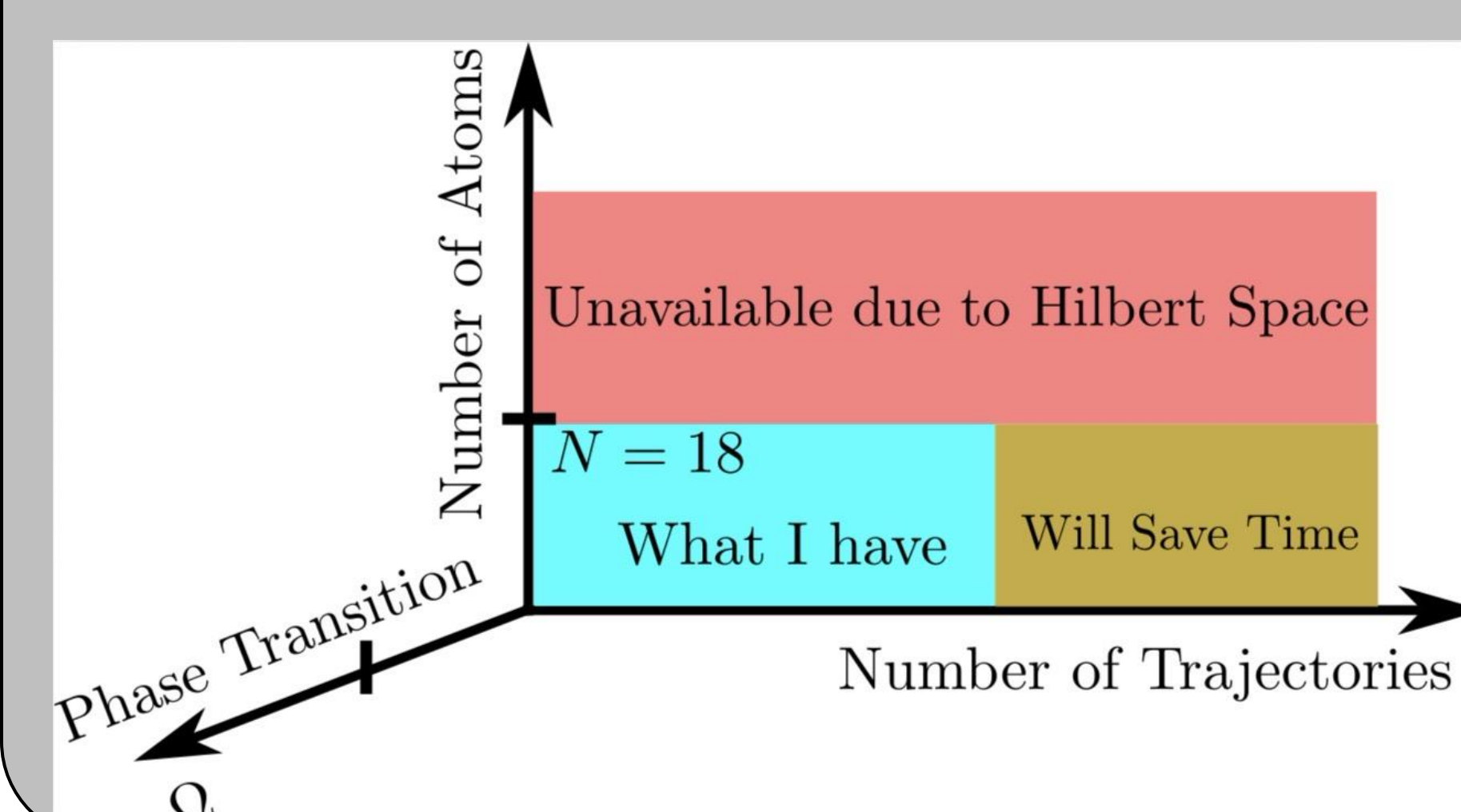
## How typical trajectory looks like (in bistability region)



4 examples of quantum trajectories  $N = 18$  (best I can get),  $\Omega = 0.73\Omega_c$  (bistability region),  $\gamma_c = 10\gamma_s$ . Red and Green lines are the mean-field solutions. Generated with QuTiP (python library for quantum simulations).

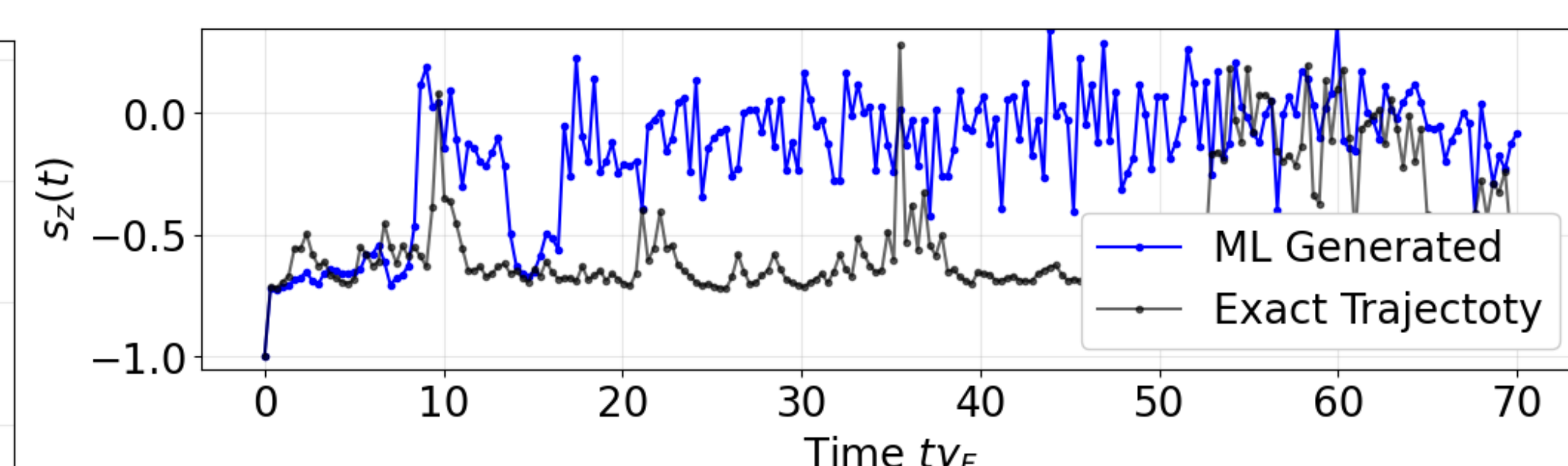
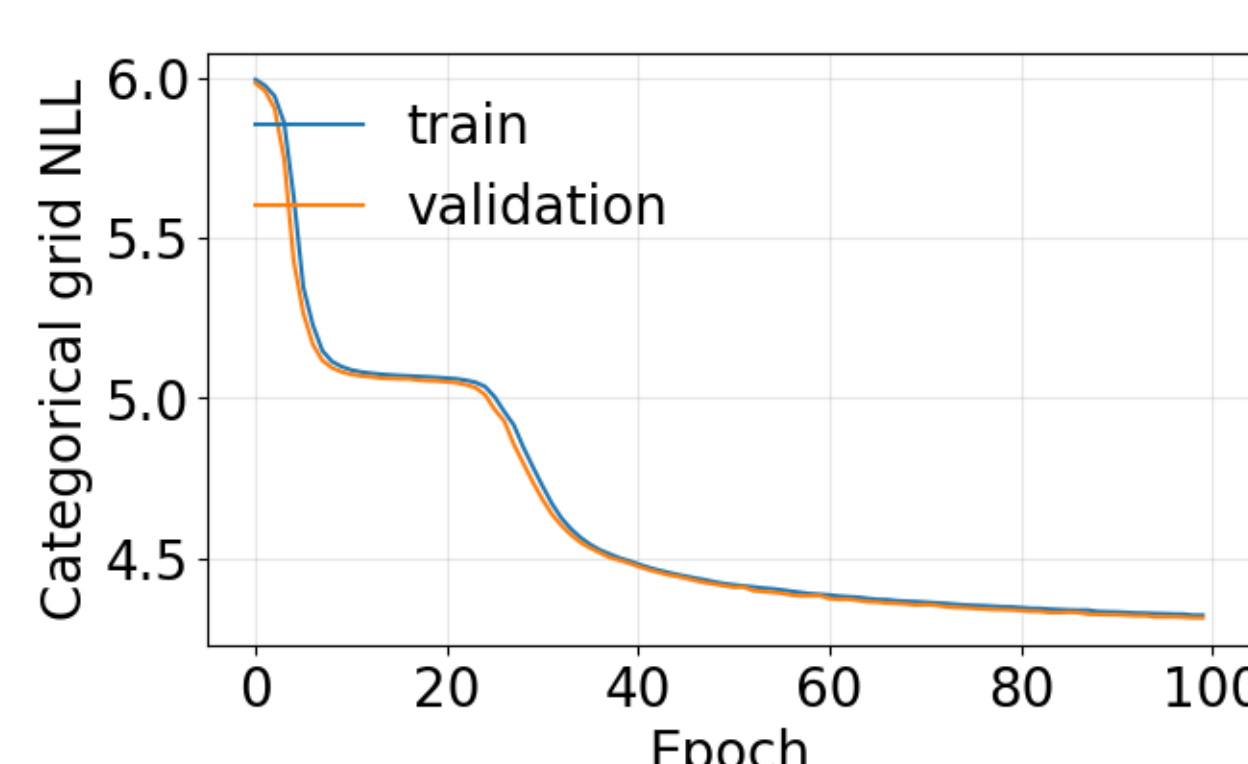
## Goal of the project

Create a model to quantum trajectories generate trajectories based on data from Monte Carlo wave-function simulations in QuTiP

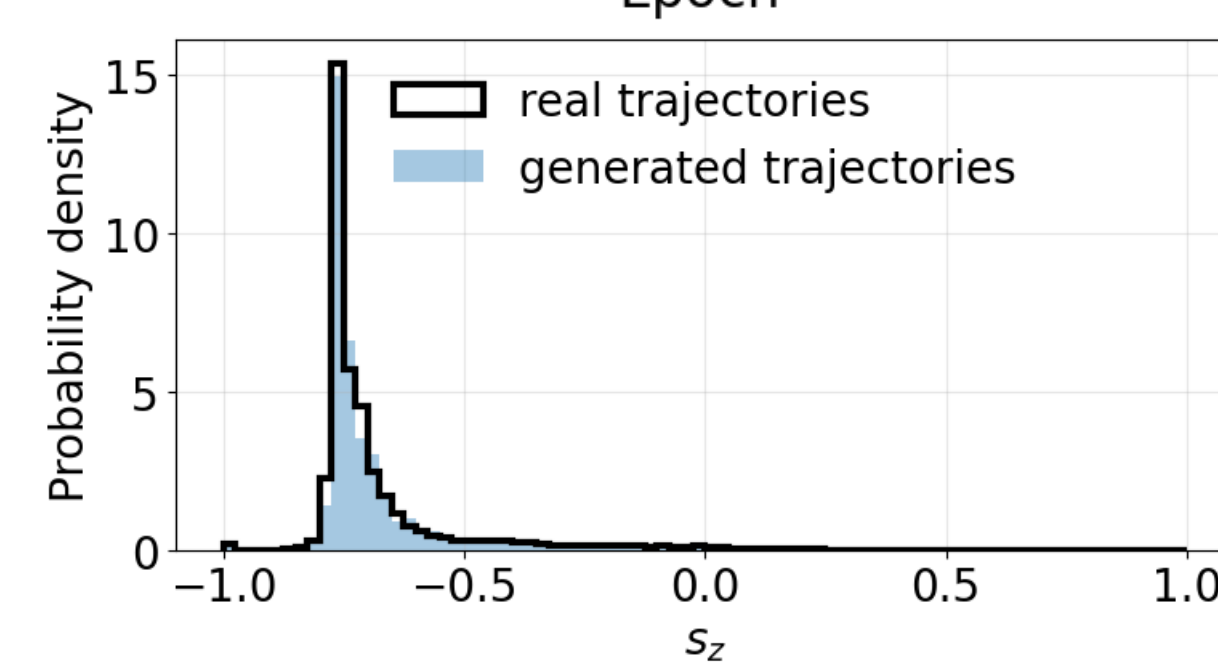


- Fix N and  $\Omega \rightarrow$  Generate More Trajectories Faster  $\rightarrow$  Capture mean and/or bistability
- Fix  $\Omega$  and try to extrapolate to unreachable N values
- Full model with N and  $\Omega$

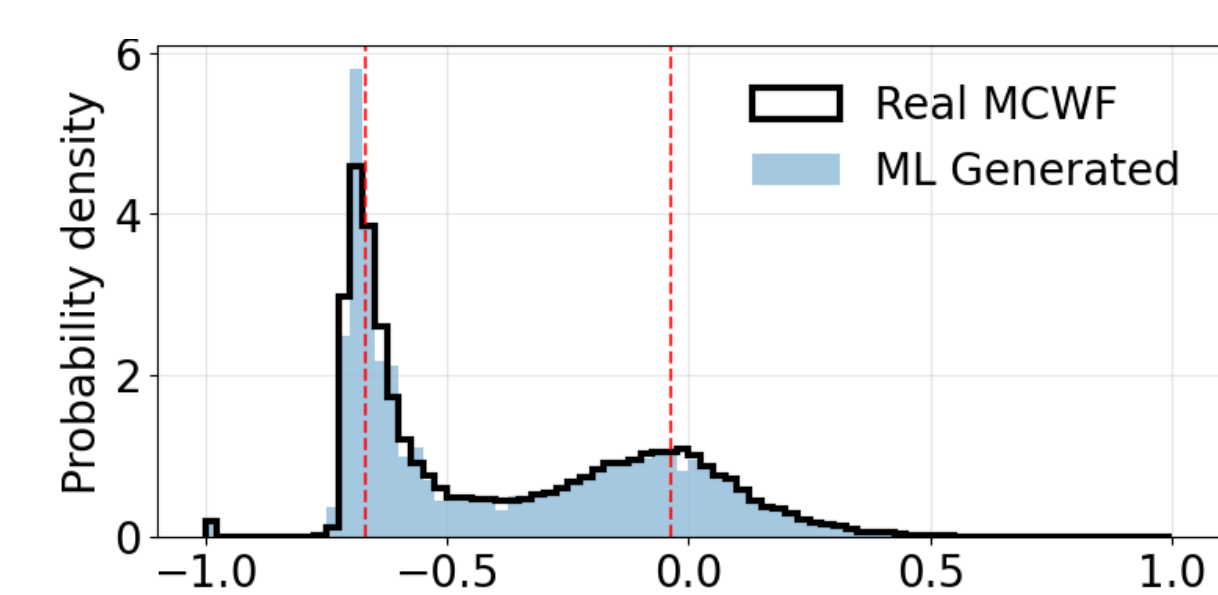
## Fixed N and Ω – test of the model



N = 18,  $\Omega = 0.73$ . Exact vs ML Generated trajectory

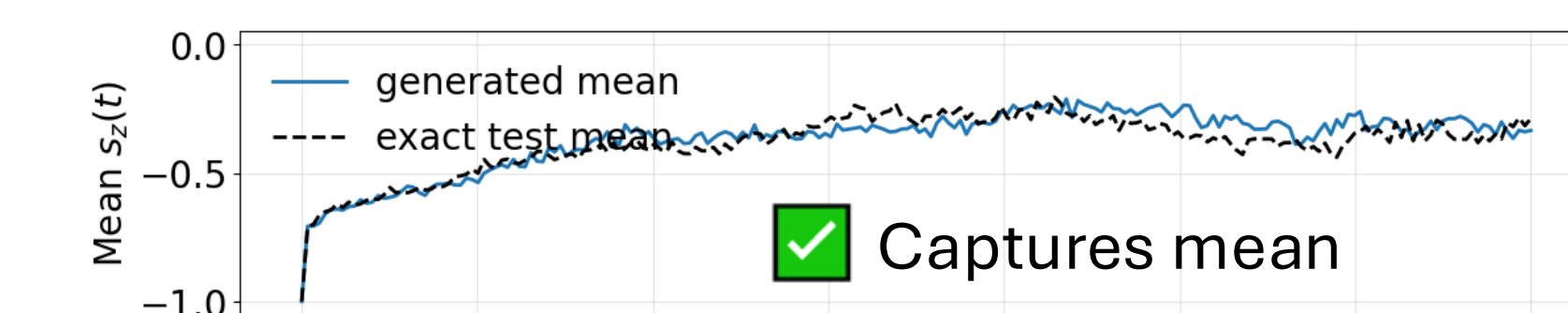


N = 8  
NO  
Bistability



N = 18  
Bistability  
Exist

✓ Captures Bistability



✓ Captures mean

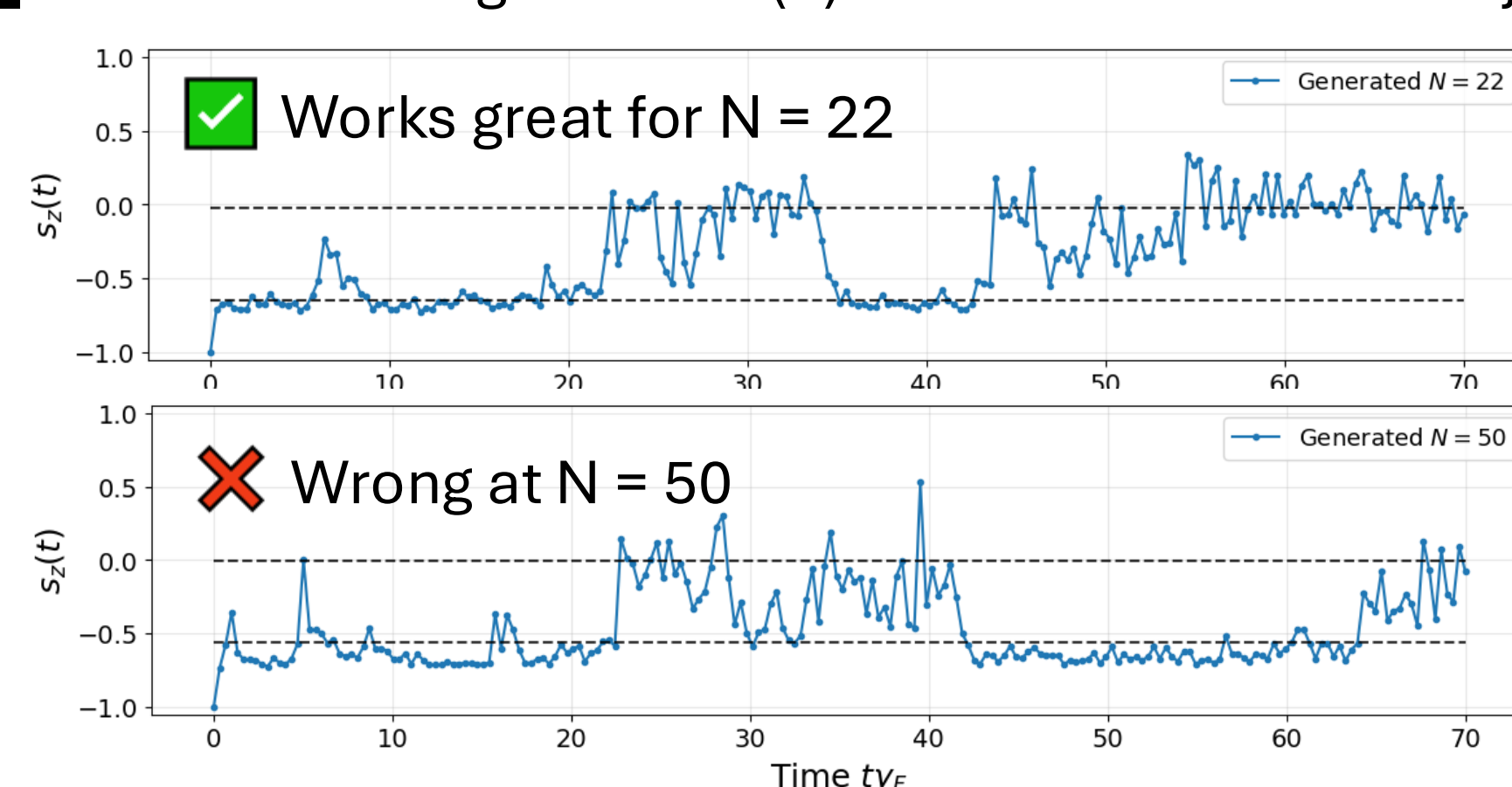


✗ Does not complete  
Real trajectory

## Fixed Ω and try to extrapolate to larger

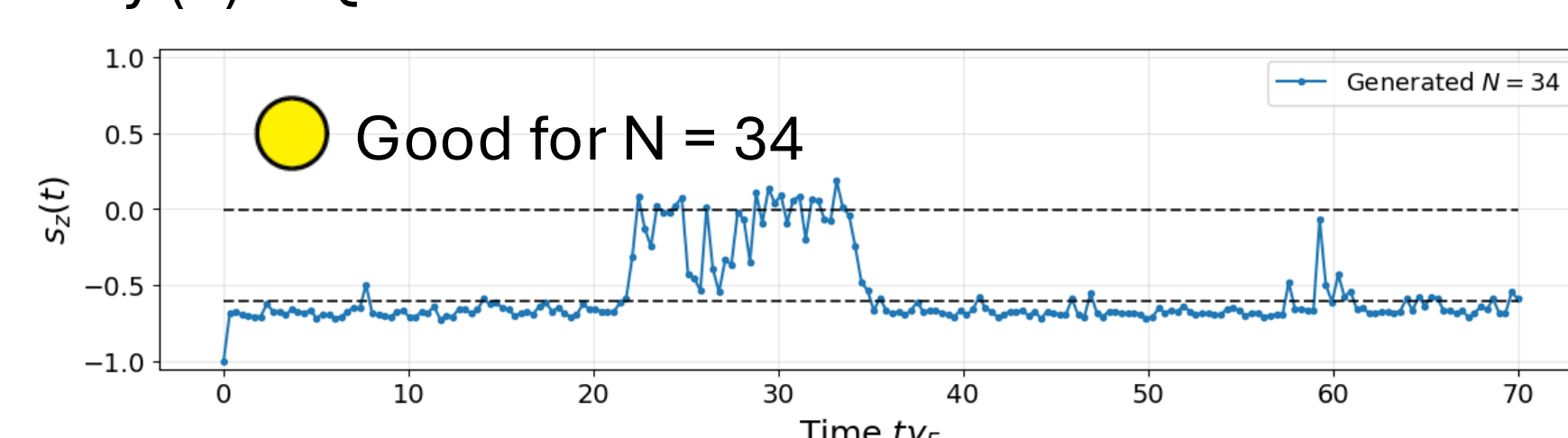
Learned on N up to 18 for  $\Omega = 0.73$  (bistability region)

I can check larger N with (a) Mean-Field for each trajectory (b) PIQS for mean value

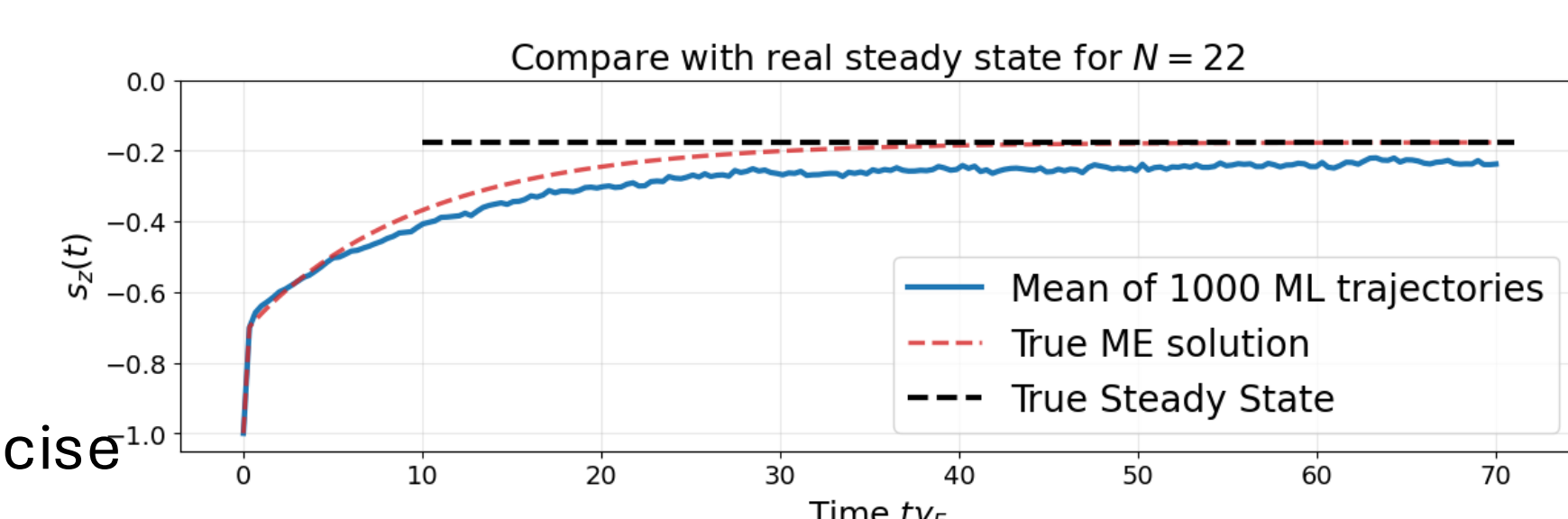


✓ Works great for N = 22

✗ Wrong at N = 50



Good for N = 34



✗ Mean estimation is not precise

## Full Model: Vary N and Ω

We take N=22 (outside of the dataset) and build for different drives (0.5, 0.6, 0.64, 0.68, 0.76, 0.8)

